

Why the Lakehouse?

INTRODUCTION TO DATABRICKS LAKEHOUSE



Gang Wang
Senior Data Scientist

What you'll learn



databricks

Explore Databricks Lakehouse

1. How data is **organized** in a lakehouse
2. How to manage **compute** and notebooks
3. How to **govern** and share data securely
4. How to **deploy** your work to production

Meet the Instructor

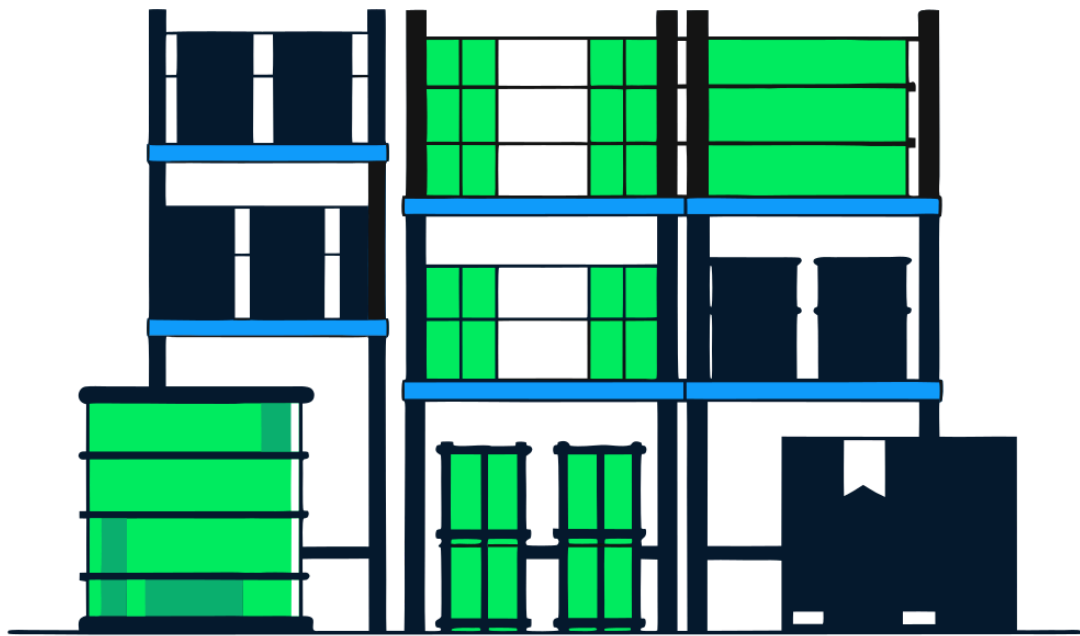
Gang Wang

- Senior Data Scientist
- Origin Energy, Australia (2021-Present)
- 9+ Years post-PhD experience



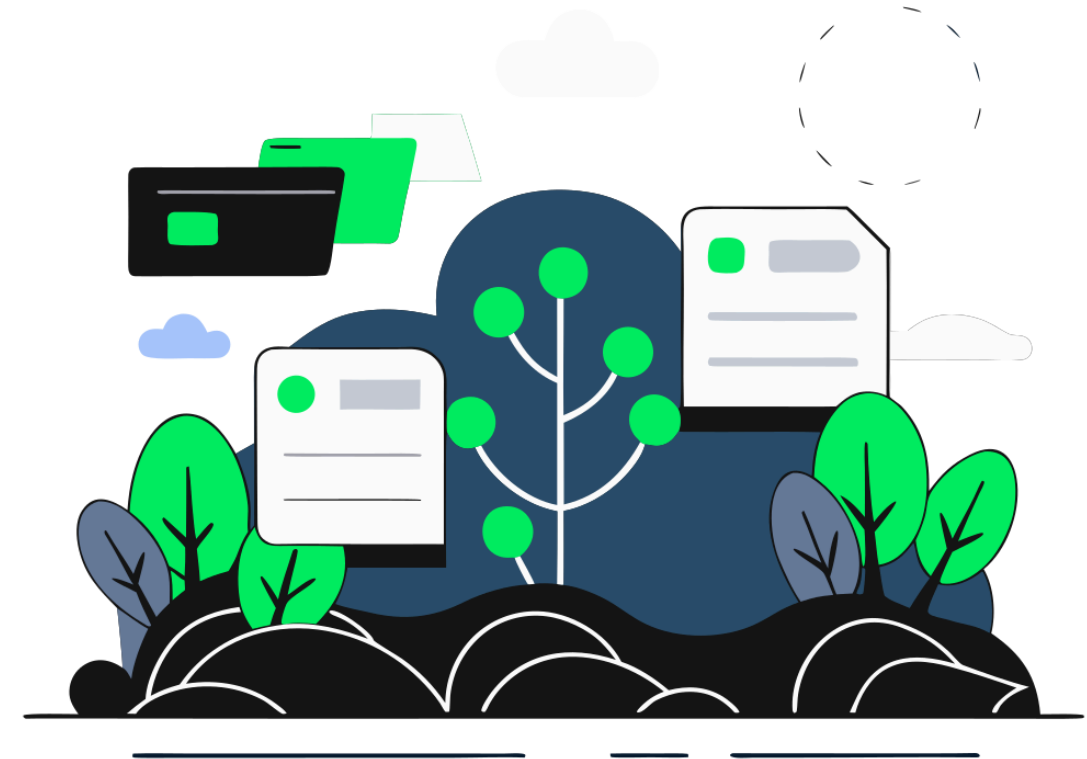
The data lake promise

- Store **any data** at low cost
- Structured, semi-structured, unstructured
- Flexible **schema-on-read**



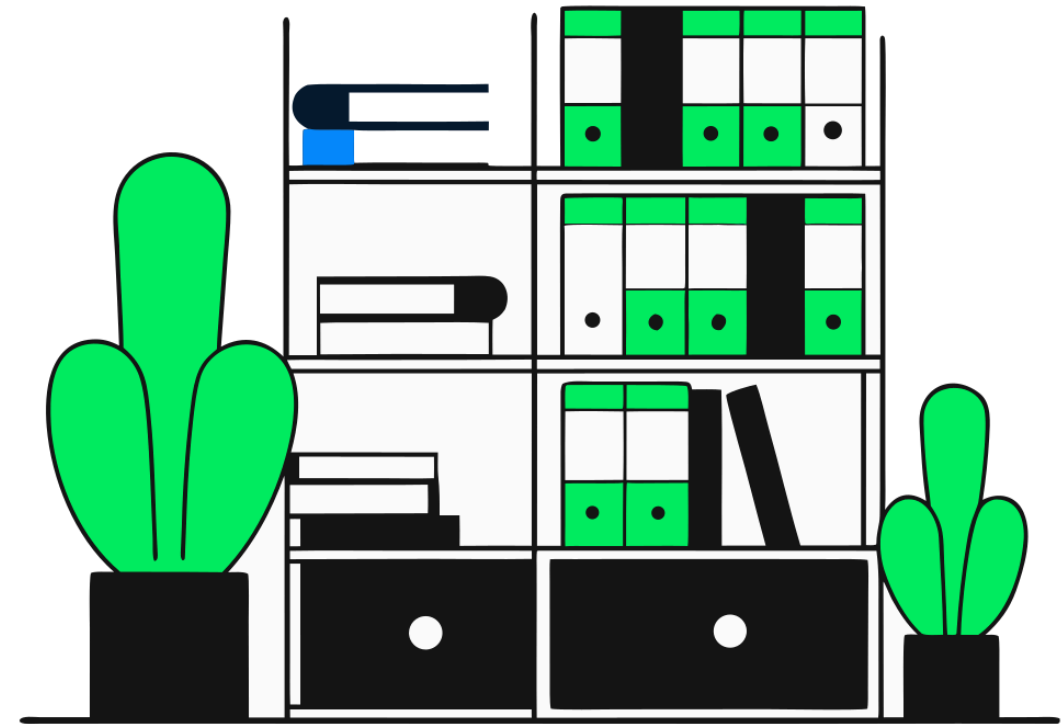
The lake's dark side

- No **guarantees** on data quality
- No built-in governance or access control
- "Data swamp" - hard to find trustworthy data
- Separate tools needed for analytics and AI

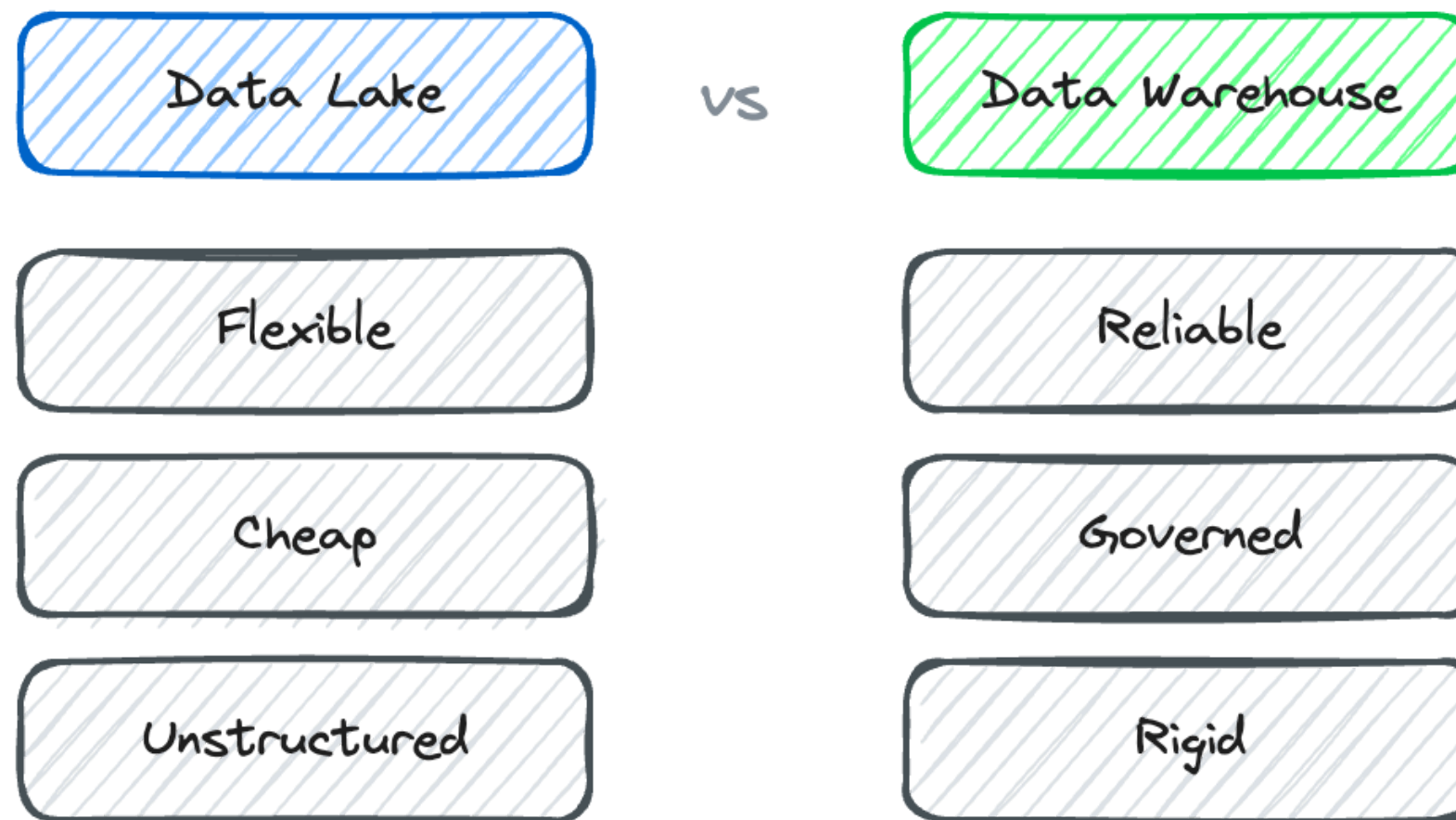


The warehouse trade-off

- **Reliable** and performant
- Strong governance
- But **expensive** and rigid
- Limited to structured data only

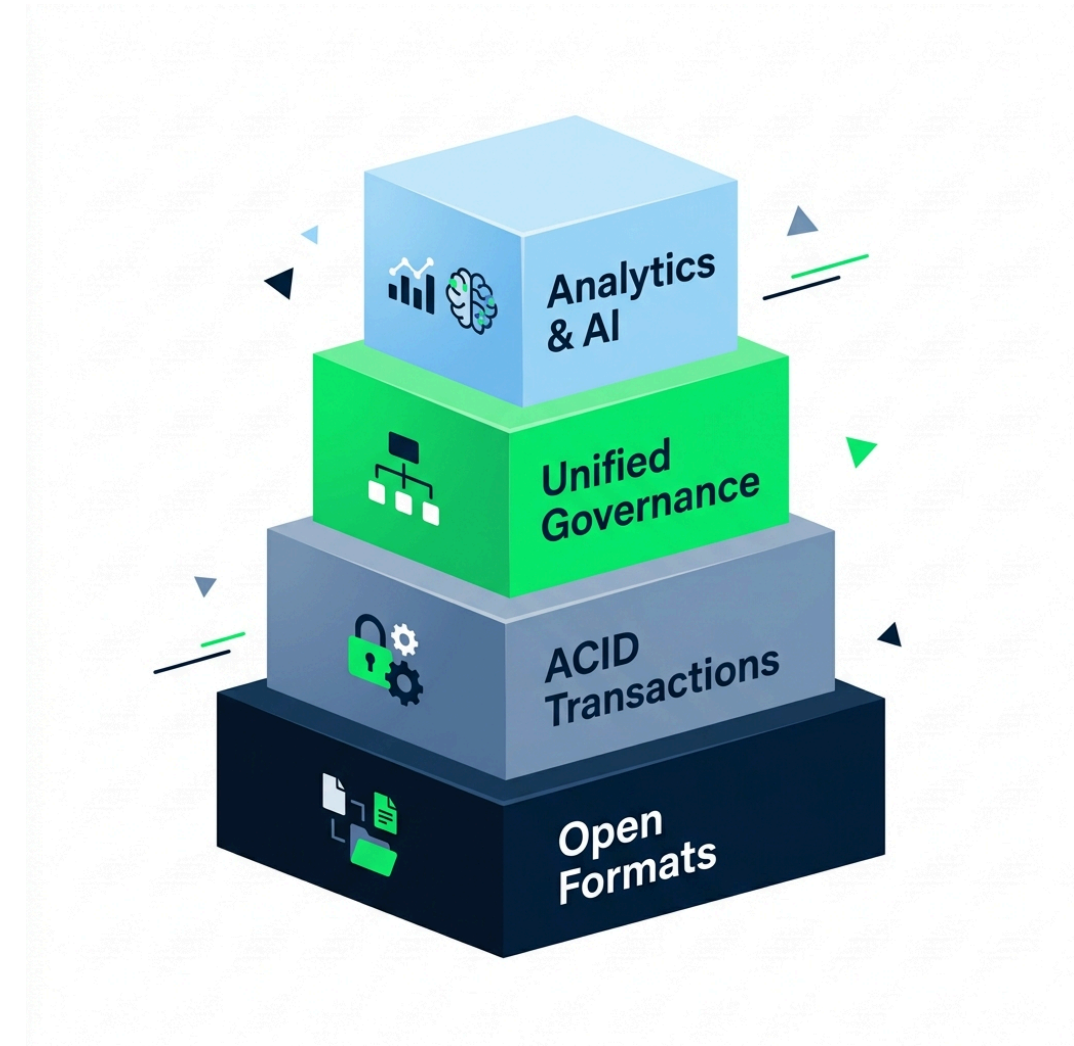


Enter the Lakehouse

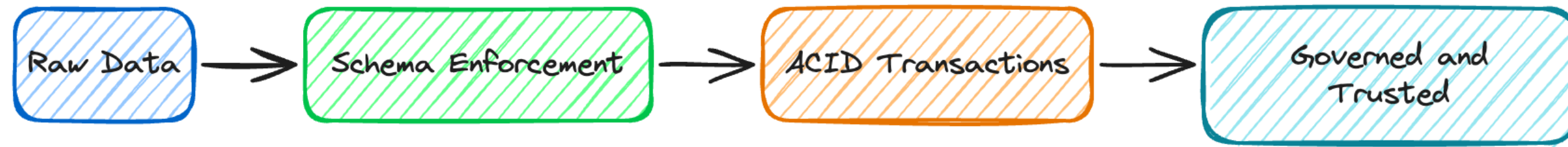


What makes it work in Databricks?

- Open file formats (Delta Lake)
- ACID transactions on the lake
- Unified governance (Unity Catalog)
- One platform for analytics, AI, and apps



Lakehouse improves data quality



- Schema enforcement at write time
- Transaction logs track every change
- Time travel - query historical versions

Summary

- **Data lakes** - flexible but unreliable
- **Data warehouses** - reliable but rigid and costly
- **Lakehouse** - combines both, one platform
- Powered by Delta Lake, ACID, and Unity Catalog

Let's practice!

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The medallion architecture

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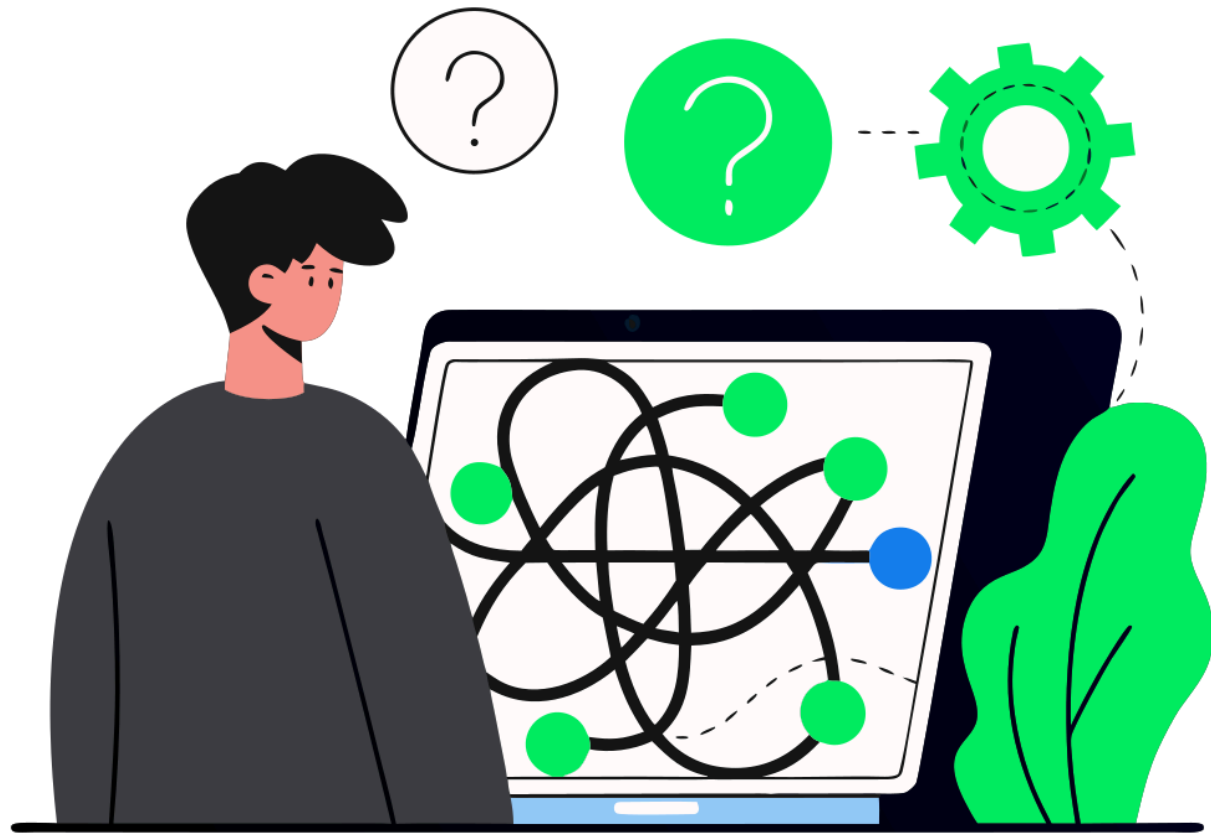


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From raw to ready

- Raw data lands in the lakehouse - **now what?**
- Messy, unvalidated records need a path to clean insights
- The **medallion architecture** solves this

The data quality problem



- Raw data from **dozens of sources**
- Missing fields, duplicates, mismatched types
- Analysts need **trustworthy data**

Three layers, one purpose



The bronze layer

- Captures data as it arrives
- Inconsistent formats and nullable fields
- **Append-only** - nothing deleted or modified
- Your **safety net** - always trace back

```
{"pickup": "2024-03-15T08:23",  
  "dropoff": "2024-03-15 8:41",  
  "fare": null,  
  "zone": "236",  
  "distance": "4.2mi"}
```

```
{"pickup": "03/15/2024 09:10",  
  "dropoff": "2024-03-15T09:32",  
  "fare": 18.50,  
  "zone": "unknown",  
  "distance": 6.1}
```

The silver layer

```
SELECT * FROM silver_taxi_trips  
LIMIT 3;
```

pickup_ts	dropoff_ts	fare	zone_id	dist_km
2024-03-15	2024-03-15	14.20	236	6.8
2024-03-15...	2024-03-15	18.50	142	9.8
2024-03-15	2024-03-15	22.00	79	12.1

- Timestamps → **proper datetime types**
- Nulls removed, duplicates eliminated
- Schema **enforced** - consistent types

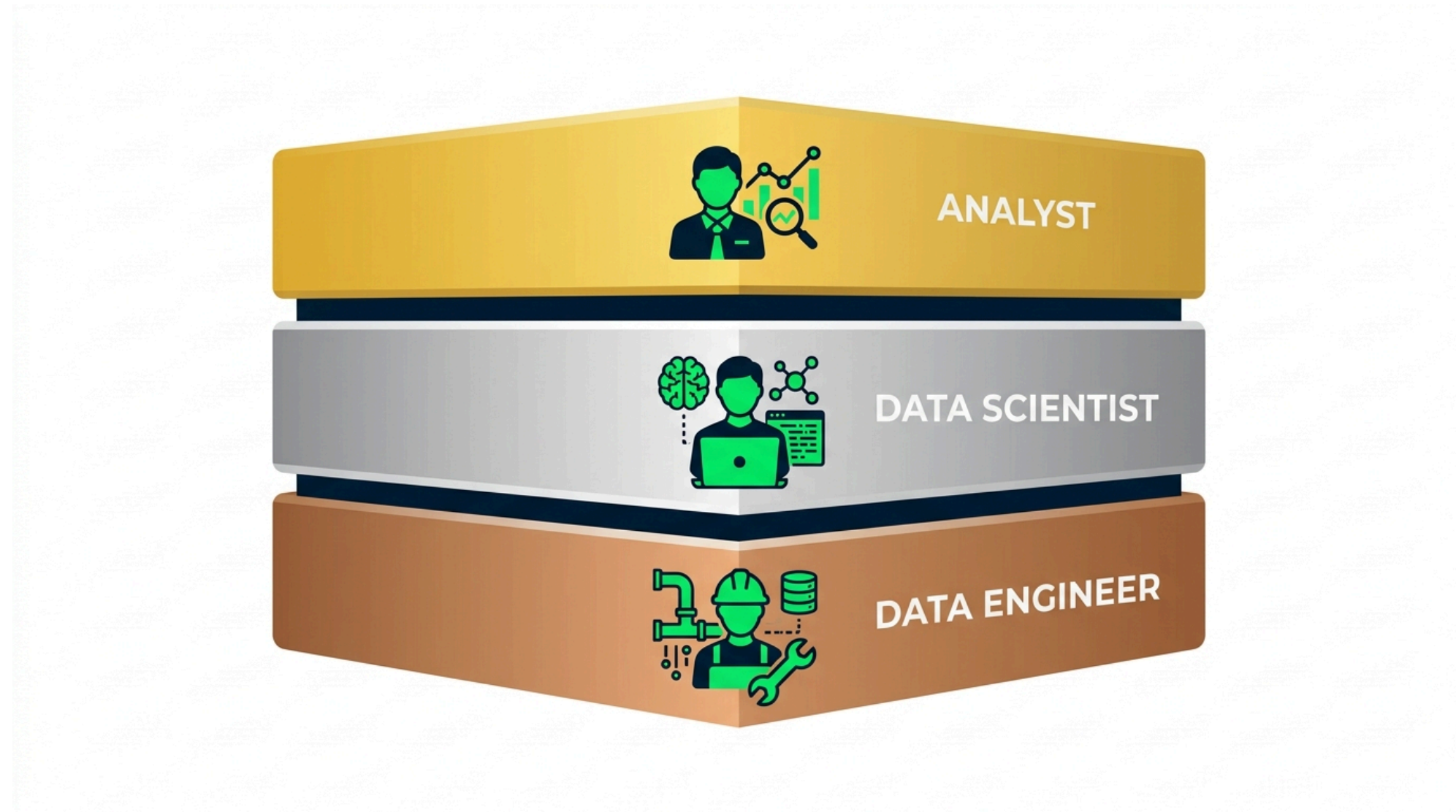
The gold layer

- Business-level aggregates
- Powers dashboards and executive reports
- Optimized for fast **consumption**

```
SELECT * FROM gold_taxi_daily  
LIMIT 3;
```

date	zone_name	avg_fare	trips
2024-03-15	Upper East	16.80	1,247
2024-03-15	Midtown	22.45	3,891
2024-03-15	Financial Dist	19.10	2,156

Who typically uses which layer?



- These are **common patterns**, not strict rules - any role may access any layer

Summary

- **Bronze** - captures everything raw, your safety net
- **Silver** - cleans and validates, reliable foundation
- **Gold** - aggregates for business, served insights
- Each layer → different consumers, different standards



Let's practice!

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Platform architecture

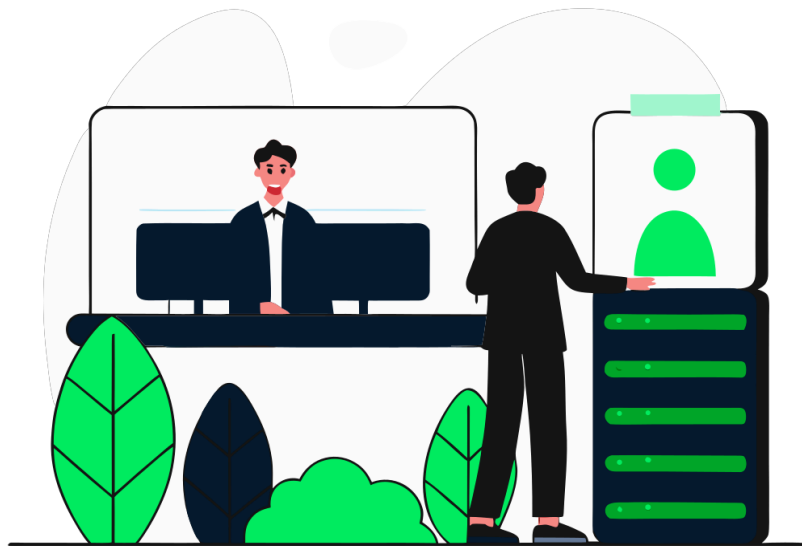
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The hotel analogy

Control plane - "The Front Desk"



- Managed by Databricks
- Handles coordination and requests

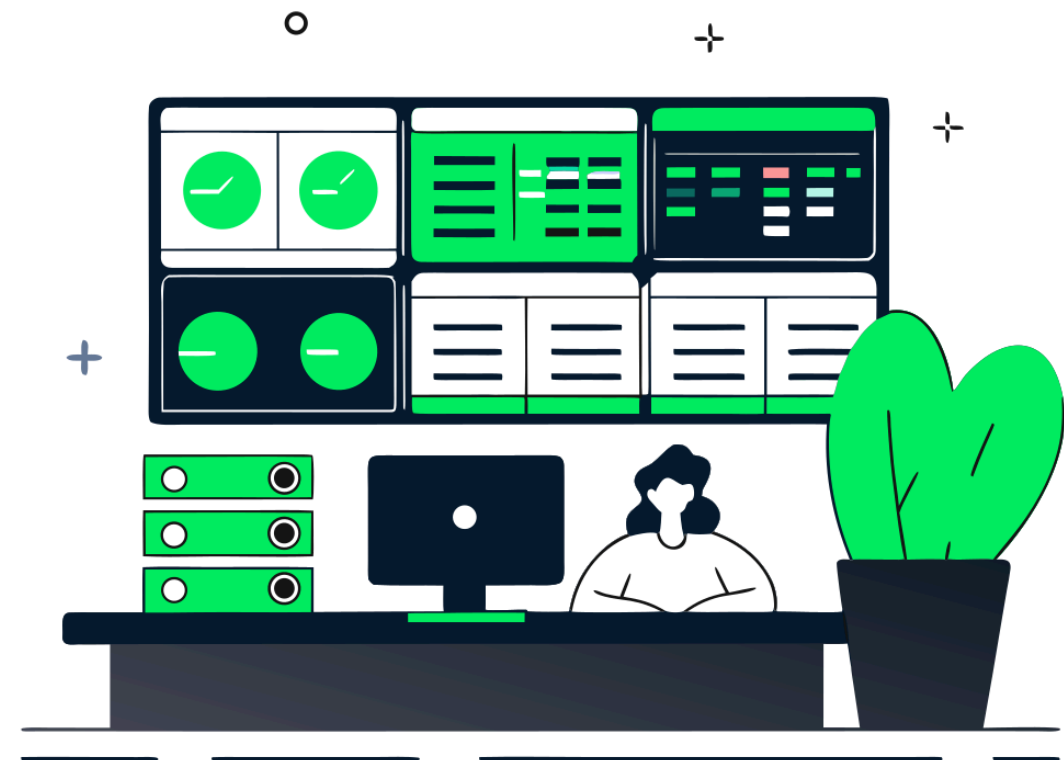
Data plane - "Your Hotel Room"



- Lives in your cloud account
- Where your data and compute live

The control plane

- Managed by Databricks
- Workspace UI and notebooks
- Job scheduling and orchestration
- Cluster management and configuration

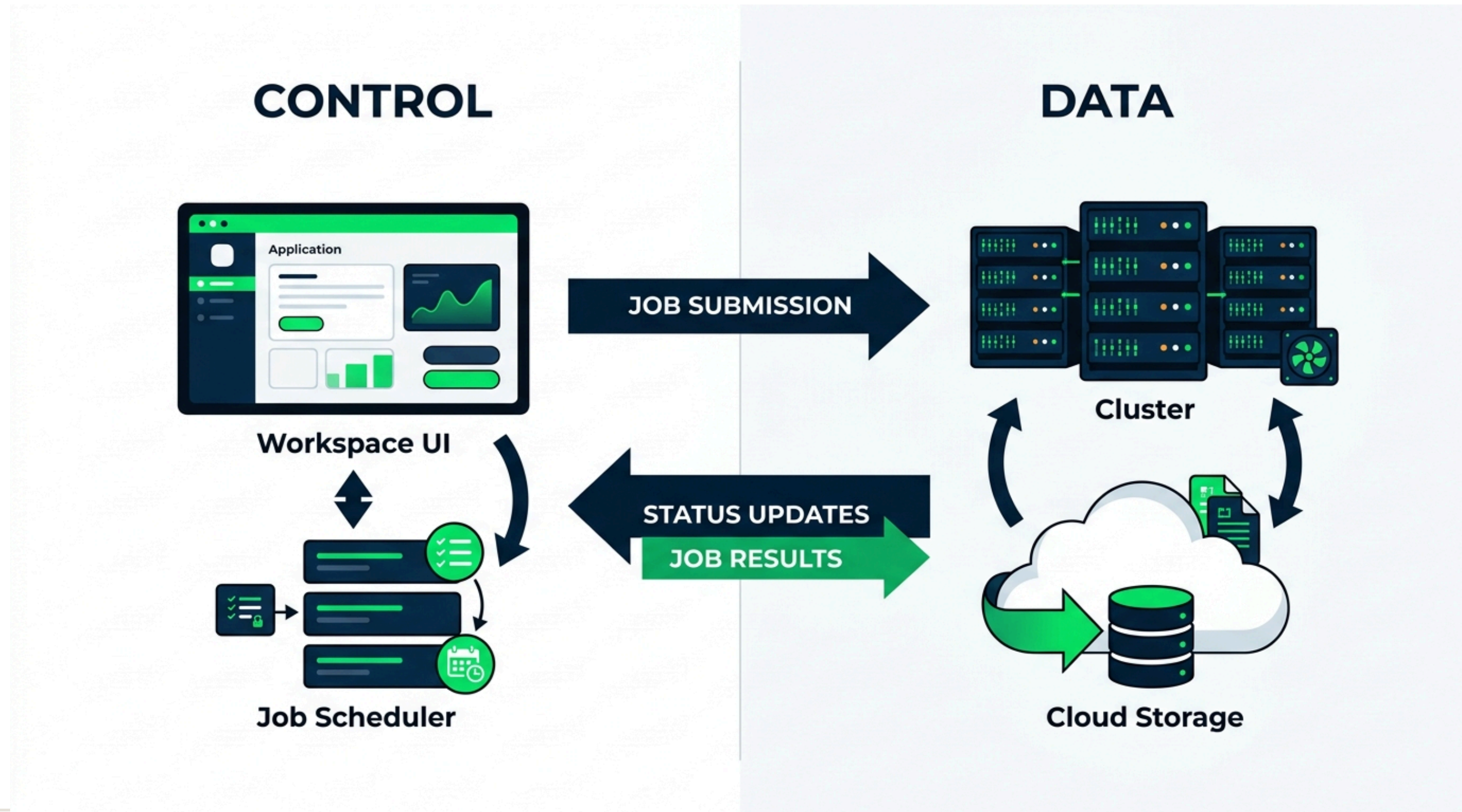


The data plane



- Lives in **your cloud account**
- Compute clusters run here
- Data stored in your cloud storage
- You control access and networking

What lives where?



Audit logs and system tables

- **Audit logs** track all workspace activity
- Stored in your cloud account as **system tables**
- Query with SQL for compliance reporting

```
SELECT user_name, action_name,  
       event_time  
FROM system.access.audit  
WHERE action_name = 'clusterCreate'  
ORDER BY event_time DESC  
LIMIT 5;
```

Summary

- **Control plane** - Databricks-managed (UI, scheduling, management)
- **Data plane** - your cloud (compute, storage, data)
- **Audit logs** - stored as system tables in your account
- Your data **never leaves** your cloud environment

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